

Warping and RDL Stress Analysis in Large Fan-Out Package with Multi-Chiplet Integration

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Abstract—As the advanced packaging technology is required to fulfill digitalized industry with big data, such as high performance computing (HPC), data center server, router, and switcher applications. Due to the hunger from above market needs, great interests were induced by both research institutes and industry. Multiple chiplets integration can provide design flexibility, high performance and power efficiency. Therefore, advanced package technology is needed for this application. Chip-last FOCoS, one of the best options, is developed as advanced packaging technology to enable different functional chiplets within a single package. This technology can improve yield, performance and cost to shorten the time to market. The package warpage control is very challenging in assembly of larger fan-out package with multi-chip integration. The re-distribution layer (RDL) trace broken risk is also key issue to influence package reliability in thermal cycling test (TCT). High CTE-mismatch effect was significantly occurred at die to die (D2D) gap. The RDL trace stress is more sensitive to material properties, process flow and geometric structure due to complex multi-layer structure in several materials. It is a tougher job to minimize RDL trace stress under each process limitation. In this paper, the RDL trace stress is compared in detail to trace layout, line/spaces (L/S), D2D gap distance and trace location at room temperature (RT) and high temperature (HT). The results indicate that the trace layout design and trace location play critical roles to enhance reliability performance.

Keywords- Multi-Chiplets integration, FOCoS, Warpage, stress, RDL (Re-Distribution Layer), D2D gap (die to die gap)

I. INTRODUCTION

In recent years, the demand of big data for digitalized industry has increased dramatically. The major application such as high-performance computing (HPC), data center server, artificial intelligence (AI), internet of thing (IoT) and 5G are developed rapidly. All these application domains require gaining and satisfying the requirements of higher design flexibility and better performance. It can be seen lately that the grown of Moore's Law driven slowly in past [1]. The semiconductor industry will go forward great integrated, functional and power efficiency architecture. Therefore, the heterogeneous chiplets integration solution is proposed to provide powerful new possibilities for IC design. The manufacture of monolithic large die is suffered limitation because of low yield and high cost. Fan-out (FO) technology as a solution that spread up fan-out chip area by epoxy molding compound (EMC) with embedded silicon die [2]. Fan-out chip on substrate (FOCoS) contains different chips connected through re-distribution (RDL) and ubumps [3-4]. Finally the fan-out chip is assembled on a high pin count ball grid array substrate through C4 bump or Cu pillar.

It integrates multiple smaller chiplets in to one functional large package. The most challenging in-process is die placement accuracy in multi-chiplets integration. The mask in lithography can't be dynamically off-set RDL pattern to cover the internal die shift. In the chip-first process, the die shift is determined to the die bonder capability. It has lower accuracy commonly due to the balance between accuracy and UPH in mass production. For the strategy of die shift improvement, fine pitch requirement, process optimization and Known Good Die (KGD) yield management, RDL-first (also called chip last) process is a candidate to effective cost reduction. FOCoS chip last (FOCoS-CL) is one of the suitable solutions that has better self-alignment accuracy (~1 μm) by the surface tension of solder in reflow procedure and high throughput in mass production [5-6]. The FO multi-chiplets structure has to be developed in chip-last platform that process flow of FOCoS-CL as shown in Figure 1. The Figure 2 shows cross-sectional image of FOCoS-CL.

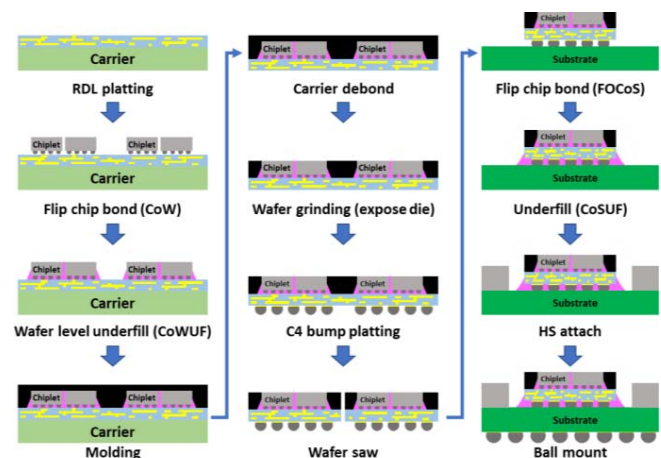


Figure 1. Illustration of process flows of FOCoS-CL.

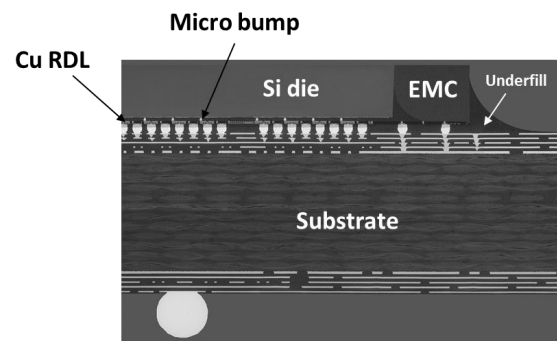


Figure 2. Cross-sectional image of FOCoS-CL.

As the size of the package becomes larger and larger, warpage control is main challenge. In this paper, a numerical model by finite element method (FEM) of fan-out chip was established to validate with Shadow Moiré measurement data. Package warpage is evaluated with different geometry and material combination. Besides, reliability performance management is also considered as critical topic for chiplets integration. High CTE-mismatch significantly occurred at D2D gap. Fine line RDL trace broken risk would be focused to investigate the mechanical stress behavior [7].

II. TEST VEHICLE AND METHODOLOGY

The 8 silicon chips are integrated to one 50x35 mm² FO die and surrounded by epoxy molding compound (EMC) with FOCoS structure as show in Figure 3. The side-by-side silicon dies are connected to side wall by chip on wafer underfill (CoWUF). The detail structure of test vehicle is listed in TABLE I. The substrate size is 80x80x2 mm³ with 6+2+6 Cu layers. The package show the larger warpage after large FO die on substrate due to the complex CTE mismatch of multi-material. The heat spreader (HS) has to be added to improve the package warpage. The both of ring and cavity type are also evaluated to several width and thickness to optimize the package warpage as shown in Figure 3(c). The package warpage is calculated to quarter model with symmetric planes at package center, and using hexagonal element with quadratic type by numerical software ANSYS.

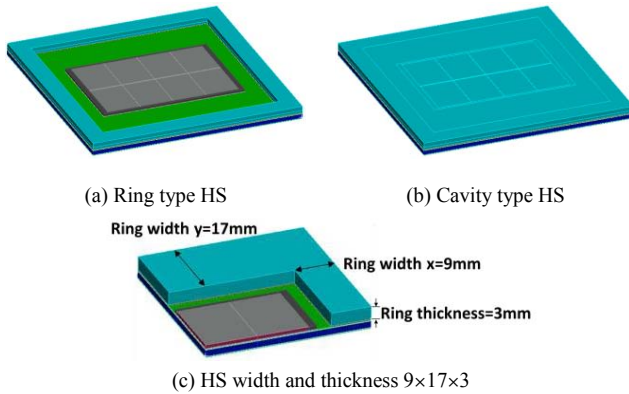
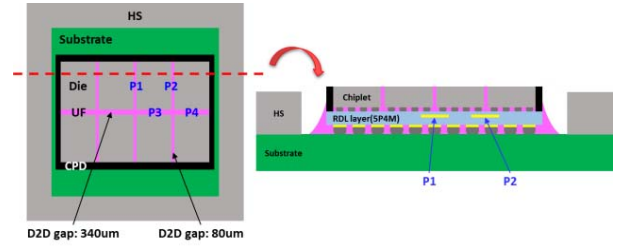


Figure 3. The package structure of multi-chiplet integration by FOCoS-CL (a) with the ring type HS and (b) with cavity type HS. (c) show the HS width and thickness.

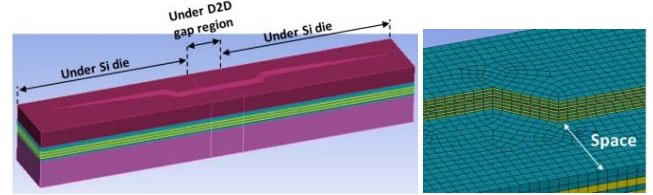
TABLE I. TEST VEHICLE INFORMATION

Package size (mm ²)	80x80			
FO die size (mm ²)	50x35			
HS Type	Ring		Cavity	
HS ring width×thickness (mm)	9×9×3	9×17×3	6×6×2.3	6×6×3
Chiplet die size (mm ²)	12×16			
RDL layer thickness (um)	4			
PA layer thickness (um)	3			
CoWUF/CoSUF thickness (um)	40/70			
Substrate thickness (um)	2100			

The interconnection between chips is through the Cu trace in 5P4M RDL layout as shown in Figure 4(a). The D2D gap is considered to 80um and 340um, and RDL L/S is 1.5/1.5um and 2/2um. The sub-model method was used to calculate the RDL stress in detail as shown in Figure 4(b). The principal stress value and direction are discussed because of the maximum principal stress dominate the crack and surface delamination of IC package. The direction of maximum principal stress is usually normal to crack surface, and the value is proportional to the probability of crack or surface delamination in-process or thermal-cycling-test (TCT). The angular and straight RDL trace layout is considered to minimize the trace stress as shown in Figure 4(c). The quadratic hexagonal element type is applied to the sub-model too, and the 75um space between maximum stress location and sub-model boundary is set to avoid that the boundary impact the stress in Saint-Venant's principle. All interfaces of materials are assumed to perfect adhesion.



(a) The RDL trace is inserted in P1 to P4 position by sub-model.



(b) The sub-model include CoWUF (dark red), CoSUF (pink), Cu layer (yellow), PA (sky blue)



(c) RDL trace for straight and angular layouts

Figure 4. (a) The sub-model is inserted in 4 positions (P1 to P4) to calculate the trace stress. The D2D gap is 80um and 340um at P1/P2 and P3/P4, respectively. (b) The sub-model is utilized to evaluate RDL stress, and the 75um space is set to avoid that the boundary impact the stress. (c) The straight and angular layouts of RDL trace are considered to multi-chiplets interconnection.

The material properties are listed in TABLE II. The thermo-mechanical simulation is based on elastic behavior, and all materials are considered to be isotropic. Temperature-dependent coefficient of thermal expansion (CTE) and modulus are considered in all calculation. The modulus, CTE and Tg of CoWUF, chip on substrate underfill (CoSUF), EMC, HS adhesive, passivation (PA) of RDL are measured

by dynamic mechanical analyzer (DMA) and thermo-mechanical analyzer (TMA), respectively.

TABLE II. THERMOMECHANICAL PROPERTIES OF THE CONSTITUENTS

Component	Modulus (GPa) @ 25°C/260°C	CTE (ppm/°C) @ 25°C/260°C	T _g (°C)
Silicon	131	2.8	-
Heat Spreader	187	17	-
Adhesive	0.1/0.02	117	220
Passivation	2/0.01	56/285	-
FO compound	28/4	8/27	170
CoWUF	10/0.6	20/80	140
CoSUF	7/0.1	30/125	90
Cu	117	16.9	-
Solder mask	3/0.3	38/155	130
Prepreg	7/0.3	23/78	150
Substrate core	25/16	8/2	300

III. SIMULATION RESULTS AND VERIFICATION

The FO die warpage was measured by Shadow Moiré from RT (30 °C) to HT (260 °C) as shown in Figure 5. The dot-blue line show the average warpage of 3 samples with error bar, and the deformation contour is plotted in blue frame. The FO die shows the concave shape at RT due to the silicon has lowest CTE 2.8 ppm/°C in all material. The PA and Cu have larger shrinkage than silicon under the cooling procedure, and the opposite behavior is shown in HT. The silicon die dominate the warpage behavior from the view of geometric structure and material modulus (ζ function) [8]. The PA and RDL are thinner than die, however its contribution of CTE in warpage is so large that the FO die warpage is very sensitive to process temperature.

The simulation results show the mirror symmetry at 260 °C due to the warpage is evaluated by the linear-elastic model. The Shadow Moiré exhibits the asymmetric profile respect to heating and cooling process. The FO die has smaller warpage in heating process than cooling one because of the several methods are applied to inhibit the in-process wafer warpage. The wafer warpage must to be met the machine handling capability in mass production. However, the external force will induce the internal stress of device to balance the deformation. When the external force is removed the residual stress leads to the warpage keep at lower region. The residual stress could be released by structure relaxation of unrestrained heating. Thus, the simulation is goodness of fit in cooling process of Shadow Moiré, and the structure relaxation start at 183°C of heating process. The maximum deviation of warpage is ~30 μ m among 3 samples in cooling process with the same temperature.

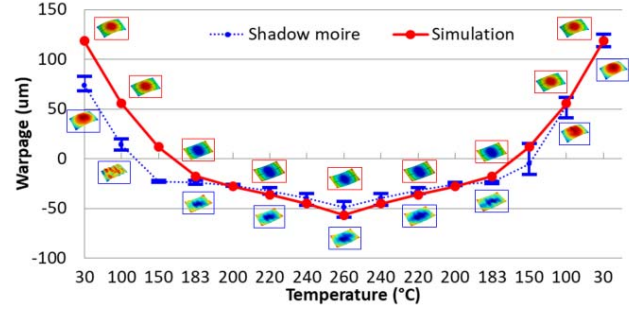


Figure 5. The circle red line shows the FO die warpage by FEM method among RT and HT. The FO die warpage is measured in detail by Shadow Moiré in dot-blue line with error bar. The contour of FO die deformation is show in red and blue frame for simulation and experiment, respectively. The warpage “-” is concave and “+” is convex.

The package warpage is also evaluated for HS type and width. The package has very large warpage after FO die on substrate at RT due to the high CTE mismatch between silicon and substrate. The package warpage has to be inhibited at RT and controlled at HT to avoid the SMT (Surface-mount technology) process fail. The higher warpage at RT lead to pick and place misalignment between solder ball and ball pad. In addition, the solder non-joint occurs in reflow process as HT warpage is over the stand-off height. Thus, the HS is utilized to package to control the package warpage. The package warpage is sensitive to HS geometric structure (ring width and thickness). The HS dominate package warpage behavior whether RT or HT, because it owns very large modulus and thickness. It is a key that choosing a suitable HS size and type to optimize the RT and HT warpage. The package warpage is calculated to several HS size and type as shown in Figure 6. The package warpage could be inhibited effectively by both HS type of ring and cavity at RT, and it drops down when increasing the ring width. The HT warpage is smaller than RT one in ring type HS, but the cavity type shows the opposite behavior.

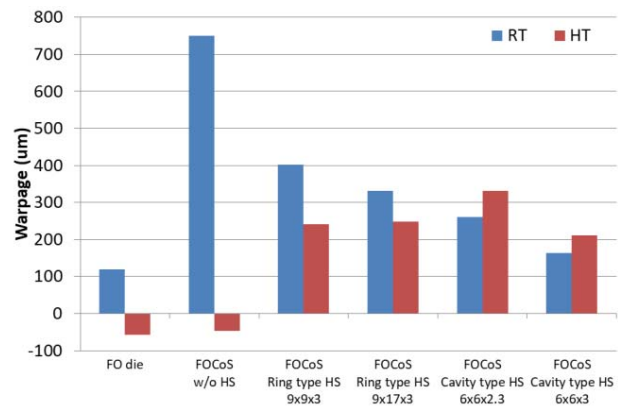


Figure 6. The FO die and package warpage is evaluated in RT and HT. The package warpage could be inhibited effectively by HS with thicker thickness and larger ring width.

The package warpage behavior determine the mass production or not, and it impact the in-process workability for example full-auto load/un-load, device quality, process

uniformity and stability. However, the device passes the reliability test or not that is determined by the suffered stress of component, interface bonding strength, material ultimate strength and material stress intensity factor and so on. In this section, RDL trace stress is discussed in detail. The trace strength is so weaker at 2um fine L/S that reducing the trace stress becomes an urgent topic.

The trace stress is dependent on the location in package. First, the trace is considered under D2D gap region and silicon die as shown in Figure 7(a). Both of trace stress are almost the same in RT, but they show dramatically difference at HT. For the P1 position, the trace occur larger stress at HT than RT at P1 position, and it is double to RT stress as shown in Figure 7(b). The maximum stress is located at the trace center that is at D2D gap center as shown in Figure 7(c). Opposite, the trace stress is smaller at HT than RT at P5 position, and it is reduced to 85%. The maximum stress is at trace corner near die edge side. The trace stress seems to be inhibited by silicon die. The root cause of differences between P1 and P5 is a tiny irregular deformation of FO die at HT as shown in Figure 8(a). The deformation could be found as the deformation zoom in 1000 times. The irregular deformation comes from the softening EMC, CoWUF and PA at HT. The trace will be bended by external force when the trace passes the D2D gap. The deformation is so tiny that it is difficult to found, however the trace stress is very sensitive to that. On the other hand, the silicon die is enough strong to avoid the irregular deformation. Thus, the trace under the silicon die could be protected by harder silicon. In addition, the FO die is show the normal and smooth deformation at RT even if the deformation zooms in 100 times as show in Figure 8(b) or 1000 times. The materials of polymer base own higher modulus when the environment temperature is under the T_g .

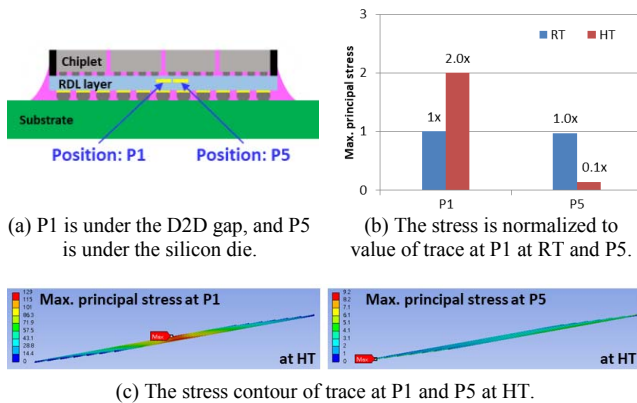


Figure 7.(a) The RDL trace is consider under the D2D gap (P1) and silicon die (P5). (b) The trace stress of straight layout with 1.5um L/S is almost the same at RT at P1 and P5. It is larger at HT than RT at P1 position, however, it is opposite at P5 position. (c) The maximum stress is located at the D2D gap at P1 position, and the maximum stress is at strace corner near die edge side.

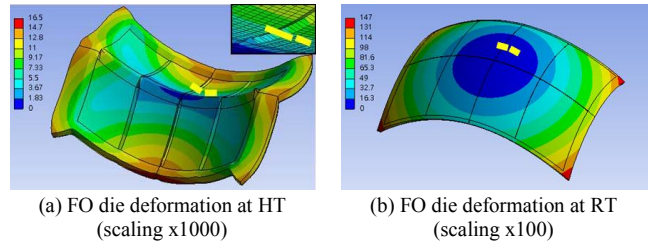


Figure 8.(a) FO die is show the normal and smooth deformation at RT and the deformation is magnified 100 times. (b) FO die has a tiny irregular deformation at HT, and it could be found as the deformation zoom in 1000 times.

The interconnection is inevitable among chiplets in FOCoS structure. The trace has to cross to the D2D gap region, thus the trace stress musts to be checked in detail to avoid the trace broken at TCT. Except for the trace location, the trace stress is sensitive to trace layout too. The angular and straight layouts are considered to minimize the trace stress. The localized stress concentration point could be transferred by the trace layout as shown in Figure 9. For straight layout the maximum stress is located at trace end at RT and D2D gap center at HT, respectively. However, it transfers to the trace angle in angular layout, and it keeps at this position as heating to HT. This method is easy control the location of maximum stress, but the side effect is larger stress at HT. The maximum stress in angular layout is double than straight layout.

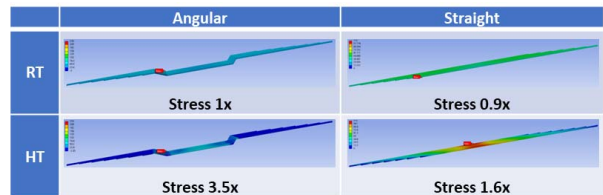


Figure 9.The contours show the maximum principal stress of RDL trace in angular and straight layout with 1.5um L/S at RT and HT, and the maximum stress value is normalized to the angular one at P2 and RT.

In this case, the stress of angular layout seems worse than straight. However, the angular layout could improve the issue of trace broken in straight layout. The key point is the trace stress under the D2D gap region. Next step, we focus on the trace stress on the D2D gap region as shown in Figure 10 due to higher risk in trace broken. The straight and angular layout is compared again at RT and HT as shown in Figure 11. At RT, the straight and angular layouts show the same stress due to package smooth deformation. The stress is increasing with heating procedure, but it is smaller in angular layout than straight one. The stress could be reduced to 34% as adopted angular layout. Although the trace angle is out-off D2D gap region and under the silicon die, the trace stress under D2D gap region could be successfully reduced by geometric structure modify. This behavior is consistent with all D2D gap location P1 to P4. The angular layout is one of solution in RDL trace broken.

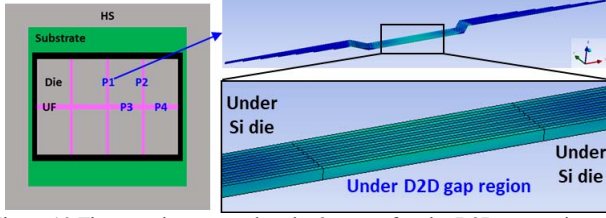


Figure 10. The trace is separated to the 2 parts of under D2D gap region and out-off D2D gap region (under silicon die).

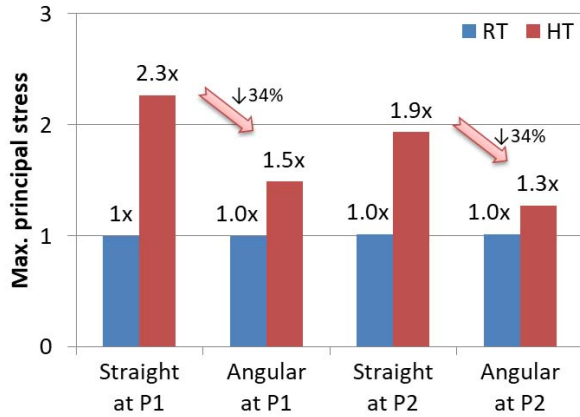


Figure 11. The trace stress under the D2D gap region is normalized to value of straight trace with 1.5um L/S at P1 at RT. The stress could be reduced 34% for angular layout comparison to the straight one at HT whether at P1 or P2.

The risk of trace broken is assessed in detail by the direction of maximum principal stress (σ_1) too. The maximum principal stress is tensile ($\sigma_1 > 0$) at RT, and it is perpendicular to the trace as shown in Figure 12. The σ_1 keeps in tensile stress as heating to HT, but its direction will rotate to parallel the trace and lay on the x-y plane. The direction is consistent with failure mode of trace broken. Concluding from the stress value and the direction of maximum principal stress, the trace broken is higher risk at HT than RT.

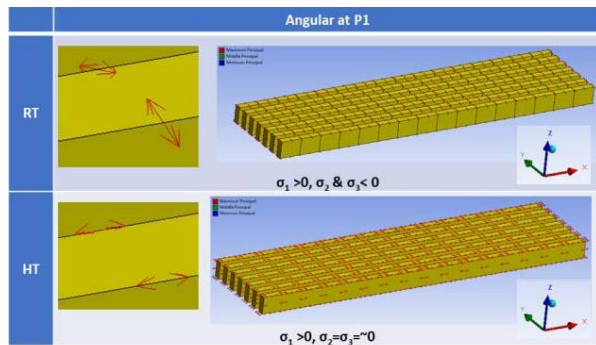


Figure 12. The direction of maximum principal stress is plotted at RT/HT and focus on the D2D gap region only. The trace locates at P1 with angular layout. The similar results are found at P2 to P4.

The stress of full trace is compared to location P1 to P4 as shown in Figure 13. The P3/P4 always occur larger stress than P1/P2 whether RT or HT. This difference comes from

the different D2D gap 80um and 340um, and it could be understand by bending effect. The lever arm is determined by D2D gap width, and the reference point is at D2D gap center. The bending moment is proportional to the lever arm. The trace angle occurs larger bending moment as enlarge D2D gap. The stresses have the same trend in straight layout and 2um L/S.

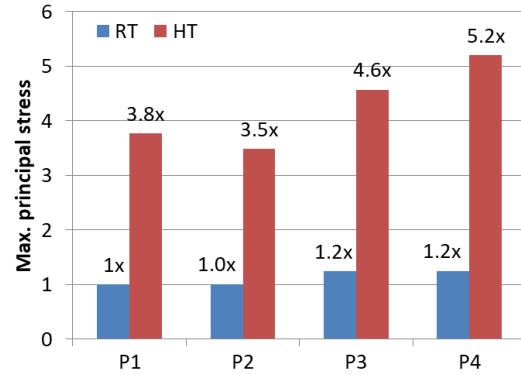


Figure 13. The trace stress is calculated to different location from P1 to P4 for angular layout with 1.5um L/S, and the P1/P2 and P3/P4 have 80um and 340um D2D gap, respectively. The P3/P4 have larger stress than P1/P2 whether RT or HT due to the D2D gap width. All stresses are normalized to the value at RT and P1.

The driving force of L/S comes from the trend in electronic device minimization and performance. The requirement of high I/O density has to arrange more trace in the RDL layer, thus the fine L/S is also evaluated as shown in Figure 14. The trace stress is decreasing at RT as reducing the trace L/S, but it is opposite at HT. The trace stress with 1.5um L/S is 1.8% to 2um L/S at HT. The reduced geometry size usually accompanies the decay of structural strength. The trace broken is higher risk at HT from previous discussion. Thus, it is more critical and challenge in assembly process and reliability test. The stress shows the similar behavior in straight layout and others locations.

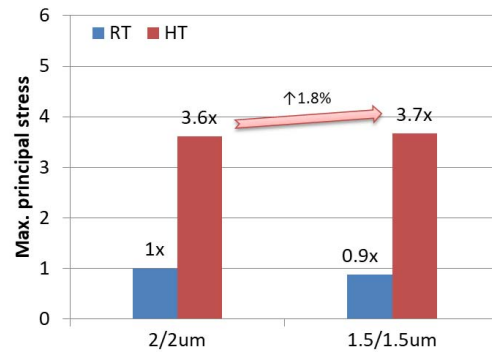


Figure 14. The trace stress is calculated to 2um and 1.5um L/S at P4, and all stresses value are normalized to stress of angular layout with 2um L/S and D2D gap 340um. The 1.5um L/S trace show the larger stress than 2um one at HT.

IV. CONCLUSIONS

The large FO package with multi-chiplet integration is studied in detail about package warpage optimization and RDL trace stress minimization. In the package warpage, the HS is used to control the package warpage. The package warpage is sensitive to HS ring width and thickness. It dominates the package warpage behavior due to large modulus and thickness. It is a key that choosing a suitable HS size and type to optimize the RT and HT warpage at same time. The FO die warpage is verified by Shadow Moiré at various environment temperatures.

The package internal residual stress could be found at heating procedure of Shadow Moiré, and it is released at ~183 °C

On the other hand, the trace broken is higher risk at HT than RT that concludes from the value and the direction of maximum principal stress. The trace is very sensitive to the package deformation under D2D gap region. The trace has larger stress at HT that comes from a tiny irregular deformation of package. The location of maximum stress is dependent on the trace layout. The trace stress under D2D gap region could be reduced to 34% as adopted angular layout. The silicon die could protect the RDL trace to avoid occur irregular deformation as all material of polymer base softening at HT. This work provides thermal-mechanical investigation for large size chiplet packages development. It also comes out some guidance for warpage control, D2D gap and RDL layout design for future FOCoS-CL.

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